

Please check that this question paper contains_ 09 questions and 02 printed pages within first ten minutes.

[Total No. of Questions: 09]

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Uni. Roll No.

29 MAY 2024

Program: B.Tech. (Batch 2018 onward)

Semester: 4th

Name of Subject: Computer Architecture and Microprocessors

Subject Code: PCIT-108

Paper ID: 16237

Time Allowed: 03 Hours

Max. Marks: 60

NOTE:

- 1) Parts A and B are compulsory
- 2) Part-C has Two Questions Q8 and Q9. Both are compulsory, but with internal choice
- 3) Any missing data may be assumed appropriately

Part – A

[Marks: 02 each]

Q1.

- a) Define word length of a microprocessor.
- b) Draw the block diagram of Stored Program Organization.
- c) What is the role of an assembler?
- d) Define arithmetic pipeline with suitable example.
- e) What is the difference between the microprocessor and microcontroller?
- f) What is the difference between cache consistency and cache coherency?

Part – B

[Marks: 04 each]

- Q2.** What is DMA? Explain the basic DMA process.
- Q3.** Write a program to perform 8 bit multiplication in 8085 processor. (Assume data and memory yourself).
- Q4.** What is flag? What are different flags available in 8085 microprocessor?
- Q5.** Write a program to add a data byte located at offset 0500H in 2000H segment to another data byte available at 0600H in the same segment and store the result at 0700H in the same segment.
- Q6.** Write the meaning of ALE, HOLD, HLDA, Logical address and Physical address.

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- Q7. Discuss the concept of pipeline and parallel processing?

Part – C

[Marks: 12 each]

- Q8. Discuss the Block diagram of 8085 in detail. Also explain its various applications.

OR

What do you mean by virtual memory? Discuss how paging helps in implementing virtual memory. Discuss any ways of improving the cache performance.

- Q9. Explain the different types of instructions available in 8085 microprocessor? Write a program to find smallest element of an array.

OR

- a) Discuss the Interfacing 8051 to LCD b) Compare High-End-High-Performance Processors Vs. Embedded Systems
